

CS 6371 - Assignment 6

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1. (a)

$$\begin{aligned}
 e &= (\text{or } (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. (a ? \text{true} : b)) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. (b ? \text{false} : \text{true})) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. ((b)(\text{false})(\text{true}))) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\text{false})(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y))
 \end{aligned} \tag{1}$$

(b)

$$\begin{aligned}
 &((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda b. (((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)(b))) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda x. \lambda y. x)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda y. (\lambda x. \lambda y. y)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda x. \lambda y. y) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda y. y) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(\lambda x. \lambda y. y)
 \end{aligned} \tag{2}$$

2. (a)

$$isempty = \lambda \ell. \text{not } (\pi_1 \ell) \tag{3}$$

(b)

$$\begin{aligned}
 head &= \lambda \ell. \pi_1 (\pi_2 \ell) \\
 tail &= \lambda \ell. \pi_2 (\pi_2 \ell)
 \end{aligned} \tag{4}$$

(c)

$$cons = \lambda e. \lambda \ell. (\text{pair true } (\text{pair } e \ell)) \tag{5}$$

(d)

$$length = Y(\lambda f. \lambda \ell. (isempty \ell ? 0_{\mathbb{N}} : succ_{\mathbb{N}}(f (tail \ell)))) \tag{6}$$

(e)

$$\begin{aligned}
rev &= Y(\lambda f.(\lambda a.\lambda \ell.(isempty \ell ? [] : f ((head \ell) :: a) (tail \ell)))) ([]) \\
append_helper &= Y(\lambda f.(\lambda \ell_1.\lambda \ell_2.(isempty \ell_2 ? \ell_1 : f ((head \ell_2) :: \ell_1) (tail \ell_2)))) \\
append &= \lambda \ell_1.\lambda \ell_2.(append_helper \ell_2 (rev \ell_1))
\end{aligned} \tag{7}$$

(f)

$$map = Y(\lambda m.\lambda a.\lambda f.\lambda \ell.(isempty \ell ? a : m ((f (head \ell)) :: a) f (tail \ell))) ([]) \tag{8}$$

(g)

$$fold_left = Y(\lambda g.\lambda f.\lambda a.\lambda \ell.(isempty \ell ? a : g (f) (f a (head \ell)) (tail \ell))) \tag{9}$$

3. (a) **Theorem.** If $e \Downarrow e''$ and $e \rightarrow_1 e'$ then $e' \Downarrow e''$.

Assume there exists a derivation D_L of $e \Downarrow e''$ and a derivation D_S of $e \rightarrow_1 e'$. We will prove by structural induction over the Small-step Semantics of Lambda Calculus that there exists a derivation D'_L of $e' \Downarrow e''$.

Inductive Hypothesis. If $e_0 \Downarrow e''_0$ has a derivation smaller than D_L and $e_0 \rightarrow_1 e'_0$ has a derivation smaller than D_S , then $e'_0 \Downarrow e''_0$.

Inductive Cases.

Case 1. Assume $D_S = (\lambda v.e_1)e_2 \rightarrow_1 e_1[e_2/v]$ and $D_L = (\lambda v.e_1)e_2 \Downarrow e''$. We must prove that $e_1[e_2/v] \Downarrow e''$.

Apply L2 to D_L :

$$\frac{\lambda v.e_1 \Downarrow \lambda v.e_1 \quad e_1[e_2/v] \Downarrow e''}{(\lambda v.e_1)e_2 \Downarrow e''} \tag{10}$$

By assumption, $e_1[e_2/v] \Downarrow e''$.

Case 2. Assume:

$$\begin{aligned}
D_S &= \frac{\frac{D_0}{e_1 \rightarrow_1 e'_1}}{e_1 e_2 \rightarrow_1 e'_1 e_2} \\
D_L &= \frac{\frac{D_1}{e_1 \Downarrow \lambda v.e} \quad \frac{D_2}{e[e_2/v] \Downarrow e''}}{e_1 e_2 \Downarrow e''}
\end{aligned} \tag{11}$$

In order to prove that $e'_1 e_2 \Downarrow e''$, it is sufficient to prove it's derivation based on L2.

$$\frac{\frac{D'_1}{e'_1 \Downarrow \lambda v.e} \quad \frac{D'_2}{e[e_2/v] \Downarrow e''}}{e'_1 e_2 \Downarrow e''} \tag{12}$$

$D'_2 = D_2$ and is thereby proved by assumption.

We apply the inductive hypothesis with $D_0 < D_S$, $D_1 < D_L$, $e_0 = e_1$, $e'_0 = e'_1$, $e''_0 = \lambda v.e$.

Therefore, $\frac{D'_1}{e'_1 \Downarrow \lambda v.e}$. Both D'_1 and D'_2 are proven, thus $e'_1 e_2 \Downarrow e''$. $\square$